t2g square lattice tight-binding derivation

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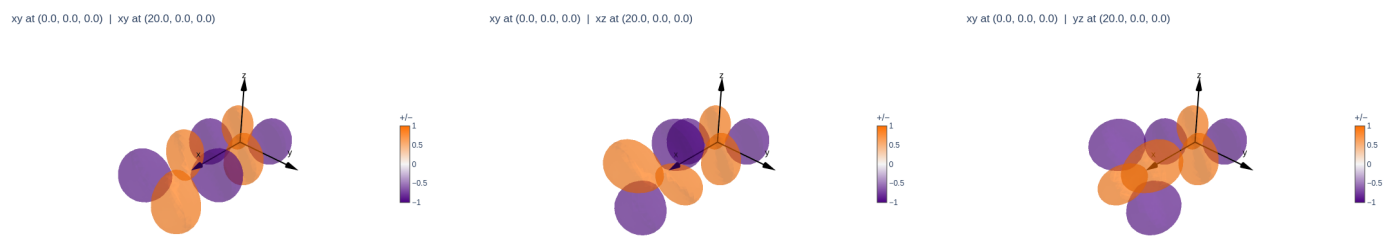
1 Single Layer

We look at hopping from all orbitals along the two x-y directions.

1.1 xy band

1.1.1 x-direction



(a) d_{xy}–d_{xy} (b) d_{xy}–d_{xz} (c) d_{xy}–d_{yz}

Figure 1: t2g orbital overlaps for nearest-neighbour hopping along x.

We look at the slater-koster integrals. The slater direction cosines are

$$l = \frac{\boldsymbol{R} \cdot \hat{x}}{|\boldsymbol{R}|} \quad m = \frac{\boldsymbol{R} \cdot \hat{y}}{|\boldsymbol{R}|} \quad n = \frac{\boldsymbol{R} \cdot \hat{z}}{|\boldsymbol{R}|} \tag{1}$$

$$\tag{2}$$

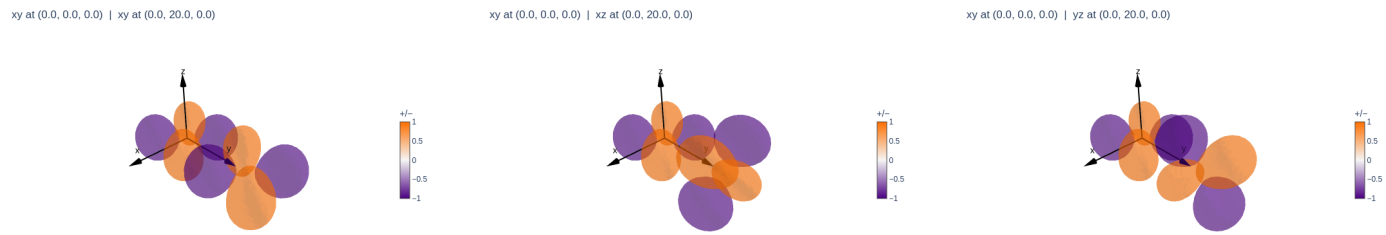
In the x direction this is just.

$$l = 1 \quad m = 0 \quad n = 0 \tag{3}$$

This simplifies the xy-xy, xy-xz, and xy-yz integrals to

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \tag{4}$$

1.1.2 y-direction



(a) d_{xy}–d_{xy} (b) d_{xy}–d_{xz} (c) d_{xy}–d_{yz}

Figure 2: t2g orbital overlaps for nearest-neighbour hopping along y.

For the y-direction the slater direction cosines are

$$l = 0 \quad m = 1 \quad n = 0 \tag{5}$$

and the integrals are

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \tag{6}$$

1.2 Tight Binding Hamiltonian

Putting this all together we get this hopping term in the hamiltonian and apply the Fourier transformation $c_i = \frac{1}{\sqrt{N}} \sum_k e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} c_{\boldsymbol{k}}$

$$\begin{aligned} H &= t \sum_{nn,\delta} c^\dagger_{i+\delta} c_i \\ &= \frac{t}{n} \sum_{nn,\delta} \sum_{q,k} e^{-i\boldsymbol{q}\cdot(\boldsymbol{R}_i+\boldsymbol{\delta})} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} c^\dagger_{\boldsymbol{q}} c_{\boldsymbol{k}} \end{aligned} \tag{7}$$